

● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

07

Q1

ADVANCED MATH

The function f is defined by $fx = 4x^{-1}$. What is the value of $f21$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$

Q2

ADVANCED MATH

In the xy -plane, the y -coordinate of the y -intercept of the graph of the function f is c . Which of the following must be equal to c ?

A. $f(0)$

B. $f(1)$

C. $f(2)$

D. $f(3)$

$$f(x) = x^2 - 18x - 360$$

If the given function f is graphed in the xy -plane, where $y = f(x)$, what is an x -intercept of the graph?

- A. $-12, 0$
- B. $-30, 0$
- C. $-360, 0$
- D. $12, 0$

A physics class is planning an experiment about a toy rocket. The equation $y = -16x^2 - 5.6x + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

Q5

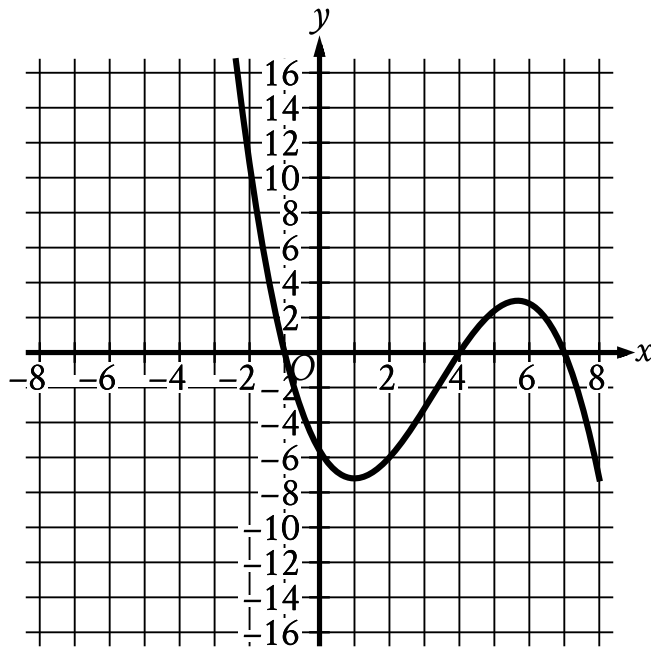
GRID-IN ADVANCED MATH

The function f is defined by $fx = 5\frac{1}{4} - x^2 + \frac{11}{4}$. What is the value of $f\frac{1}{4}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- In quadrant 2 the curve falls sharply to cross the x axis at point (negative 1 comma 0).
- In quadrant 3 the curve falls sharply to cross the y axis at point (0 comma negative 5.5).
- In quadrant 4:
 - The curve falls gradually to a relative minimum at point (1 comma negative 7).
 - The curve rises sharply to cross the x axis at point (4 comma 0).
- In quadrant 1:
 - The curve rises sharply to a relative maximum at point (5.5 comma 3).
 - The curve falls sharply to cross the x axis at point (7 comma 0).
- In quadrant 4 the curve falls sharply.

The graph of $y = f(x)$ is shown, where the function f is defined by

$f(x) = ax^3 + bx^2 + cx + d$ and a , b , c , and d are constants. For how many values of x

does $fx = 0$?

- A. One
- B. Two
- C. Three
- D. Four

Q7

ADVANCED MATH

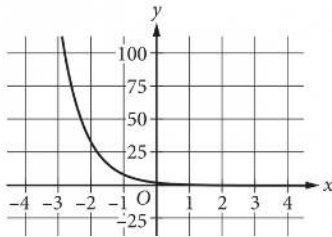
A rectangle has a length of x units and a width of $x - 15$ units. If the rectangle has an area of 76 square units, what is the value of x ?

- A. 4
- B. 19
- C. 23
- D. 76

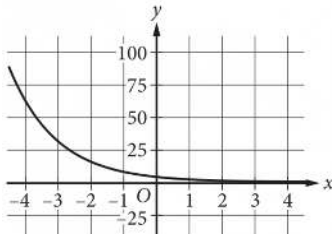
$$y = 4(2^x)$$

Which of the following is the graph in the xy -plane of the given equation?

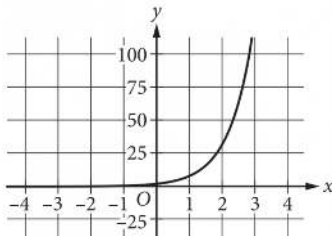
A.



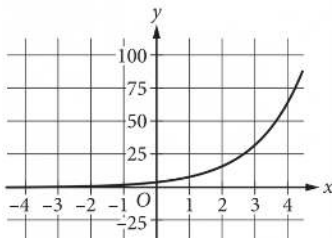
B.



C.



D.



Q9

ADVANCED MATH

The height, in feet, of an object x seconds after it is thrown straight up in the air can be modeled by the function $h(x) = -16x^2 + 20x + 5$. Based on the model, which of the following statements best interprets the equation $h(1.4) = 1.64$?

- A. The height of the object 1.4 seconds after being thrown straight up in the air is 1.64 feet.
- B. The height of the object 1.64 seconds after being thrown straight up in the air is 1.4 feet.
- C. The height of the object 1.64 seconds after being thrown straight up in the air is approximately 1.4 times as great as its initial height.
- D. The speed of the object 1.4 seconds after being thrown straight up in the air is approximately 1.64 feet per second.

A company opens an account with an initial balance of \$ 36,100.00. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A , where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is \$ 68,071.93. Which equation could define A ?

A. $A(t) = 36,100.001.05^t$

B. $A(t) = 31,971.931.05^t$

C. $A(t) = 31,971.930.05^t$

D. $A(t) = 36,100.000.05^t$